

Cones of Dunshire (Part II)

30 Points Possible

10/9/2025

Attempt 1



In Progress

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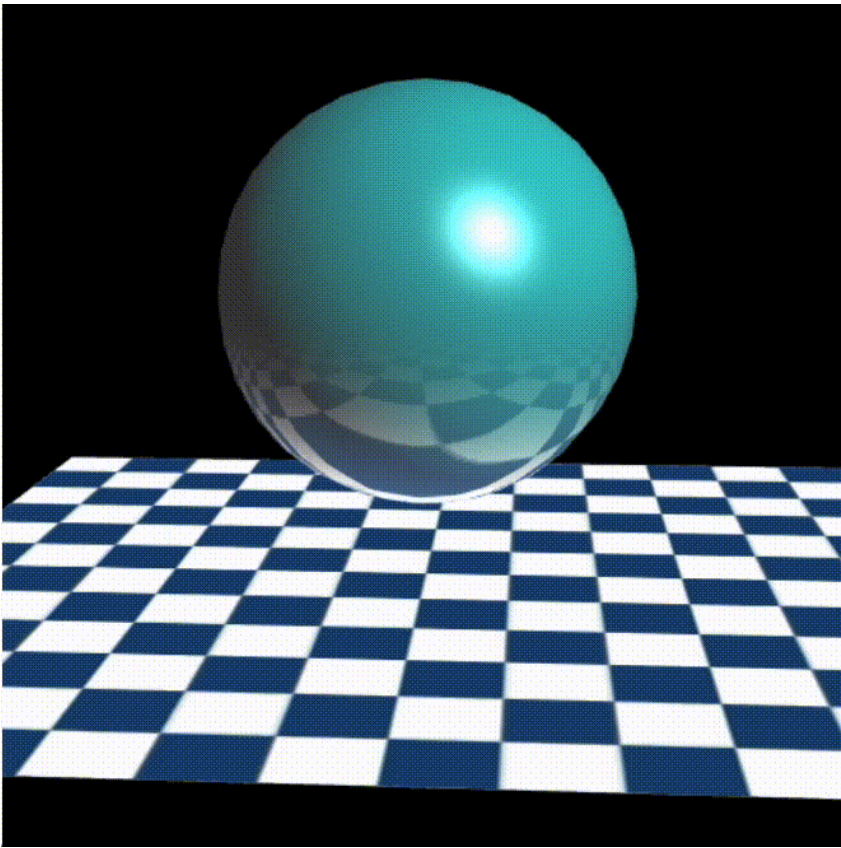
Unlimited Attempts Allowed

12/12/2025

▼ Details

A several days due-date extension will be granted to students that attend class on Thursday, Oct 9th.

Synopsis: Update the first part of the assignment with more advanced lighting effects, textures, and new controls.



Requirements: Like before, your scene should have at least three shapes present: a sphere, a cone, and a cylinder.

1 (+10 points) Have there be ambient, diffuse, and specular lighting to the scene. Have at least two lights which may be point lights or directional lights. In order for the lights to function correctly, you *must* ensure that translation is not applied to any Normal vectors. This line must be in your vertex shader.

```
v_Normal = vec3(u_ModelMatrix * vec4(a_Normal.xyz,0.0));
```

(By setting the 4th variable of 'a_Normal' to 0.0, the entire 4th row of u_ModelMatrix will be zero'd out when the two matrices are multiplied. That row is usually used to handle translations. Thus, only rotation and scale will be

applied to the normals.)

2 (+5 points). Add a Plane to the scene that acts as a "floor", similar to what you did in Windmill. The floor must have a texture and that texture may not come from any of my (Oleksiy's) website examples. Do not forget that you *must* add "normals" to your "initPlane" function since your GLSL code requires normal attributes! Ensure your texture has dimensions that are a power of 2 so that there are no potential issues loading it. The website <https://www.base64-image.de/>  (<https://www.base64-image.de/>) is a good way to load in the image as a string. You might refer to [this floor texture example](https://www.ecst.csuchico.edu/~oal-saadi/Graphics/source/anisotropic.html) (<https://www.ecst.csuchico.edu/~oal-saadi/Graphics/source/anisotropic.html>)

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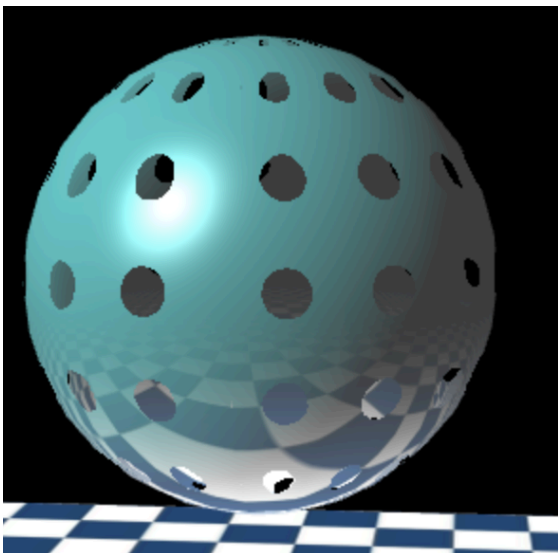
3 (+5 points) Use Reflection mapping floor to reflect the floor off of *two or more* of your shapes. You may need to turn a cone or cylinder "sideways" so the reflection is evident. That also means that your normals for those objects should be set up correctly! Take the following GLSL code to add a reflection to your shapes:

```
vec3 V = normalize(v_Position.xyz - Camera); //Camera
should be a vec3 holding its position.
vec3 reflectview = normalize(reflect(V,normal));
float u = -(v_Position.y + 2.0) / reflectview.y;
vec2 reflmapresult = v_Position.xz - u*reflectview.xz;
vec4 texcolor = texture2D(u_Sampler, reflmapresult/16.0) *
```

```
pow(max(-normal.y,0.0),2.0);
```

The code will only work if you have `"v_Position = u_ModelMatrix * a_Position;"` in the vertex shader so that all of the transformations have been applied in `v_Position`. Make modifications to the code as necessary to create a nice final image (what happens if you get rid of `"pow(max(-normal.y...))"`). Of course, add `texcolor` to the value of `gl_FragColor`.

4. (+5 points). Add a procedural texture to one of your objects from [this link](https://thebookofshaders.com/examples/?chapter=09)  (<https://thebookofshaders.com/examples/?chapter=09>). Be smart about how you add it so that it works with your existing code; you might replace all instances of `"u_resolution"` with `"vec2(1.0,1.0)"` as a starting point. Instances of `"gl_FragCoord"` should likely be replaced with `"v_UV"` (i.e. your texture coordinates). After applying the texture, use `"discard;"` in GLSL to remove all of the solid white (or black) fragments. If done correctly, you will have something that you can see through.



5 (+5 points). Update the Camera Movement so that the user has complete freedom to both walk and look in any direction. Here is one way that this can be accomplished: allow the user to use the w and s keys to move forward and back, respectively. Then, have the arrow keys cause the Camera "lookAt" position to change. So, if the user presses $\uparrow$, then the camera lookAt position will move upwards. Similarly, $\leftarrow$, $\downarrow$, and $\rightarrow$ will move the lookAt position in those respective directions. The lookAt position of the camera must change (like in Windmill, but now the user can look up or down) and may not simply be set to (0,0,0) like it was in Oasis.

Hint: You must ensure that the specular light functions correctly even when the camera position is changing! You will have to pass camera position using uniforms to GLSL since camera position is used in the specular light calculation. Camera position is also used in the reflection calculation.

Restrictions: This assignment can be accomplished using the Matsuda libraries that are provided. If you begin with [starter.html code \(link\)](https://www.ecst.csuchico.edu/~oal-saadi/Graphics/source/starter.txt) (<https://www.ecst.csuchico.edu/~oal-saadi/Graphics/source/starter.txt>) then these libraries will be loaded in for you. If you want to use additional libraries, you may request me (the instructor) to confirm they retain the low-level complexity we are studying. It remains that submitting AI-generated code is considered an ethics violation!

Turn-in: You may upload your file with a .html extension if you embedded your

Javascript into the HTML. If you split up your Javascript and HTML into separate files then you must upload a zip file having all necessary files. Your assignment must be easily run by simply opening your .html file in the browser without any additional modification, or it will receive a 0.

Grading: See the rubric below for grading criteria.

View Rubric

Cones of Dunshire (Part II)			
Criteria	Ratings		Points
Three Shapes and Phong Lighting	Sufficient Both the Cylinder, Cone, and Sphere are visible and present and have the full Phong lighting model applied. 10 pts	Not met The shapes are not visible or some part of the Phong model is missing. 0 pts	/10 pts
Plane	Sufficient There is a Plane with normals correctly applied! It is textured too. 5 pts	Not met There is no plane. 0 pts	/5 pts
Reflection Mapping	Sufficient A reflection is mapped to one of the shapes. 5 pts	Not met Reflection does not work. 0 pts	/5 pts

Criteria	Ratings		Points
Procedural Texture and Discard	Sufficient There is a procedural texture mapped using UV's and "discard;" to throw away some color to make the shape "see-through" 5 pts	Not met 0 pts	/5 pts
Updated Camera	Sufficient The user has freedom to move around the scene and change the camera's "lookAt" position, including looking up and down. 5 pts	Not met Not complete. 0 pts	/5 pts

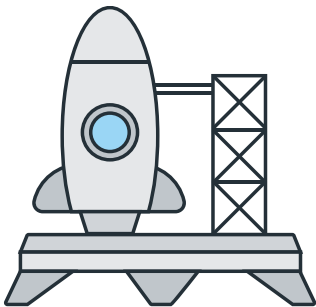
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